



TailorTalk

Your Mission ■

Build a **conversational AI agent** that can assist users in searching, filtering, and discovering files within a designated Google Drive. The agent should be capable of engaging in a natural, back-and-forth conversation with the user, understanding their search intent, executing accurate queries to find specific files (by name, type, content, or date), and presenting the results seamlessly through chat.

Technical Stack Requirements ■

- **Backend:** Python with FastAPI
- **Agent Framework:** LangGraph or LangChain
- **Frontend:** Build the chat interface using **Streamlit**
- **LLM/Chat Model:** Use any LLM API like **Gemini**, **Groq**, **OpenAI**, etc., to power the natural language understanding.

Google Drive Integration Instructions

- Use a Service Account for integrating with the Google Drive API.
- **Setup:** A sample Google Drive folder has been provided with a mix of file types (PDFs, Docs, Sheets, images) for you to use as the test repository. **Make a copy of this folder** into your own Google Drive, then share it with your Service Account's email address. The agent will search within this folder.
- **Discoverability Focus:** Your implementation must utilize the `files.list` method and heavily leverage the Drive API's `q` (query) parameter to ensure highly accurate search results. The agent should understand how to search by exact name, partial name (`contains`), file type (`mimeType`), and text within the document (`fullText`).

■ Sample Folder (copy this to your Drive):

<https://drive.google.com/drive/folders/1qkx58doSeYrcLjHPDysJyVJ36PsSqqlt>

Your Final Submission

- **Host both the backend and frontend** using a platform like **Railway**, **Render**, **Fly.io**, etc.
- Your final submission must include a working **Streamlit URL**, where I will test the file discovery functionality live.
- Include the **GitHub** code link.
- Ensure the bot is fully functional, conversational, and accurately retrieves files based on complex user requests.

***Hint:** Use the function calling / tool calling feature in LangChain to create a dedicated DriveSearchTool. The LLM should be prompted to translate the user's natural language request (e.g., "Find the financial report from last week") into a properly formatted Google Drive query parameter string (e.g., name contains 'financial' and modifiedTime > '2023-10-24T12:00:00') to pass into your tool!*

*** Let me know if you'd like to adjust the difficulty or add any requirements regarding returning download links or file previews!